DATA STRUCTURES CA

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Student Information

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Chosen Theme

Mobility Innovation:

Proposals may include, but are not limited to solutions that:

- Pilot and showcase new mobility solutions in the Docklands, combining emerging technologies with sustainable, people-centred transport, and measure uptake and equity.
- Expand active travel initiatives by using data and community input to improve walking, cycling, accessibility, and safety, and assess their effectiveness.
- Explore responsible drone applications for logistics, monitoring, and public services, ensuring regulation, safety, and trust are built in – and tested from the outset.
- Partner with industry, academia, and civic groups to co-develop scalable mobility projects and engage residents in shaping and evaluating the future of urban transport.

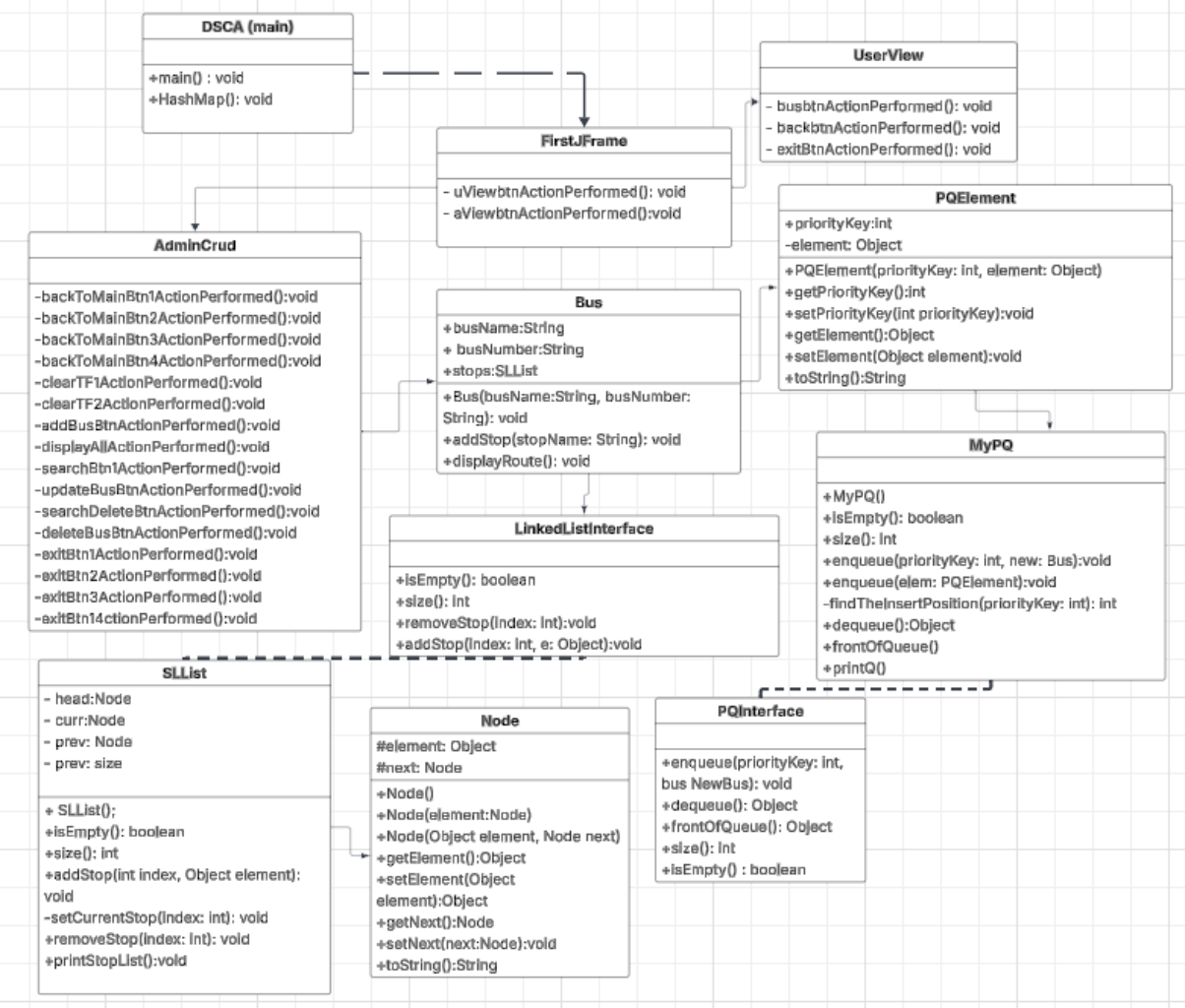
Application Description

This application aims to improve bus transport options within the Docklands by providing users with clear and accessible route information. Using public available data, it helps commuters plan journeys efficiently and encourages the use of public transport, reducing traffic congestion and carbon emissions.

These bus routes are implemented into the programme using singly linked lists, priority queues and hash maps. Singly linked lists represent the sequence of stops on a route, priority queues show the users which bus will arrive first, and hash maps allow users to quickly search for specific bus stops. This programme will allow for future updates, including real-time service information.

Administrators have full CRUD (Create, Read, Update, Delete) access to manage bus routes, while regular users can explore routes through the applications' interface.

How it's solving the problem in relation to the theme.

The project is designed to improve bus transport options in the Docklands. The application addresses this by providing a clear and accessible platform for users that displays bus routes using publicly available data. Users can easily select a bus route and access information such as its stops. By focusing on buses, the application helps users better understand available public transport options, enabling more efficient journey planning and encouraging the use of public buses. This approach can help reduce traffic congestion in the docklands while also improving accessibility for residents, commuters and visitors/tourists.

Class Diagram